

Research on Background Removal Methods for Portrait and Product Images and Development of a Web Service Based on BiRefNet

1. Introduction and Relevance

Automatic background removal from images (background removal / image matting) is one of the fundamental tasks of computer vision, widely used in video conferencing, e-commerce, media production, and data preprocessing for machine learning systems.

In the context of e-commerce, background removal becomes critically important: marketplaces require uniform product card design with a white or transparent background. Automating this process reduces operational costs for sellers and shortens time-to-market for new products. Additionally, background removal is essential for creating product catalogs, generating marketing materials, and ensuring visual consistency across platforms.

2. Review of Methods and State of the Art (SOTA)

2.1 Classical Methods

Before the advent of deep learning, the task was addressed using color segmentation and graph-based methods:

- **GrabCut (Rother et al., 2004)** — iterative graph optimization based on Gaussian mixture models of colors; requires manual annotation of a region of interest.
- **Chroma Key** — removal by color range in HSV space; applicable only with a chromatic background (green/blue screen).

The fundamental limitation of classical methods is their inability to correctly process semi-transparent areas (hair, fur) and complex textured backgrounds.

2.2 Deep Learning-Based Methods (2020–2024)

Year	Model	Architecture	Key Feature
2020	U2-Net	U-Net + nested blocks	First lightweight matting model; used in rembg until 2023
2020	Background Matting	ResNet-50 + background capture	Requires a reference frame of the clean background
2021	Background Matting V2	ResNet-50 + MobileNetV3	Real-time on GPU; improved hair processing
2021	Robust Video Matting (RVM)	MobileNetV3 + recurrent decoder	Specialized for video streams
2022	InSPyReNet	Swin Transformer / ResNet	Pyramid features; strong results on portraits
2022	DIS (Dichotomous IS)	IS-Net	First method for highly detailed object segmentation
2023	RMBG-1.4 (BRIA AI)	IS-Net fine-tuned	Commercial quality; non-commercial license

2024	BiRefNet	Swin-L + bilateral references	Achieves state-of-the-art performance on DIS5K, HRSOD, and UHRSD benchmarks
2025	RMBG-2.0 (BRIA AI)	BiRefNet + ViT / Transformer	Improved BRIA; commercial license

2.3 Current SOTA — BiRefNet (2024)

BiRefNet (Bilateral Reference for High-Resolution Dichotomous Image Segmentation, Zheng et al., 2024) is considered one of the current state-of-the-art open-source models for high-quality background removal and image matting.

Key architectural features:

- Backbone: Swin Transformer Large (hierarchical features with local attention)
- Bilateral Reference: the model explicitly models both foreground and background as complementary signals
- Multi-scale decoder: restores details at the individual pixel level (hair, eyelashes)

Results on standard benchmarks:

Benchmark	Metric	BiRefNet	Previous SOTA
DIS5K	MAE ↓	0.021	0.031 (IS-Net)
DIS5K	F-measure ↑	0.876	0.817 (IS-Net)
HRSOD	MAE ↓	0.018	0.024 (InSPyReNet)
UHRSD	S-measure ↑	0.921	0.893 (InSPyReNet)

3. Model Selection and Justification

3.1 Selection Criteria

The following practical constraints were considered when selecting the model:

1. Platform: Apple Silicon M1 Pro (ARM, no CUDA)
2. Real-time requirement: not required — offline preprocessing task
3. Edge quality: priority (hair, small details, product contours)
4. License: permissive for research and commercial use

3.2 Finalist Comparison

Criterion	MediaPipe	BiRefNet	SAM 2 (Meta)	RMBG-2.0 (Bria)
Apple Silicon CoreML / MPS	✅ native	✅ ONNX+CoreML	⚠️ partial (MPS)	⚠️ partial (MPS/need export)
Open license	✅ Apache 2.0	✅ Apache 2.0	✅ Apache 2.0	❌ commercial

Final choice: BiRefNet. For this implementation, the BiRefNet model was used through the rembg library, which provides a convenient ONNX-based inference interface.

At the time of implementation, newer models such as RMBG-2.0 were either commercially licensed or lacked stable deployment support via ONNX Runtime + CoreML on Apple Silicon. Therefore, BiRefNet was selected as the best compromise between quality, reproducibility, and deployment simplicity.

4. Implementation

4.1 Solution Architecture

A web service was developed using the following stack:

FastAPI → rembg (BiRefNet ONNX) → PIL (post-processing) → PNG (response)

- FastAPI — asynchronous Python framework; provides REST API and built-in HTML interface
- rembg — provides a convenient ONNX-based inference interface for the BiRefNet model
- ONNX Runtime with CoreML Execution Provider — hardware acceleration on Apple Silicon
- PIL/Pillow — input image conversion and RGBA PNG export

4.2 Key Technical Decisions

Session initialization at startup (not per request):

```
```python
session = new_session("birefnet-general")
```
```

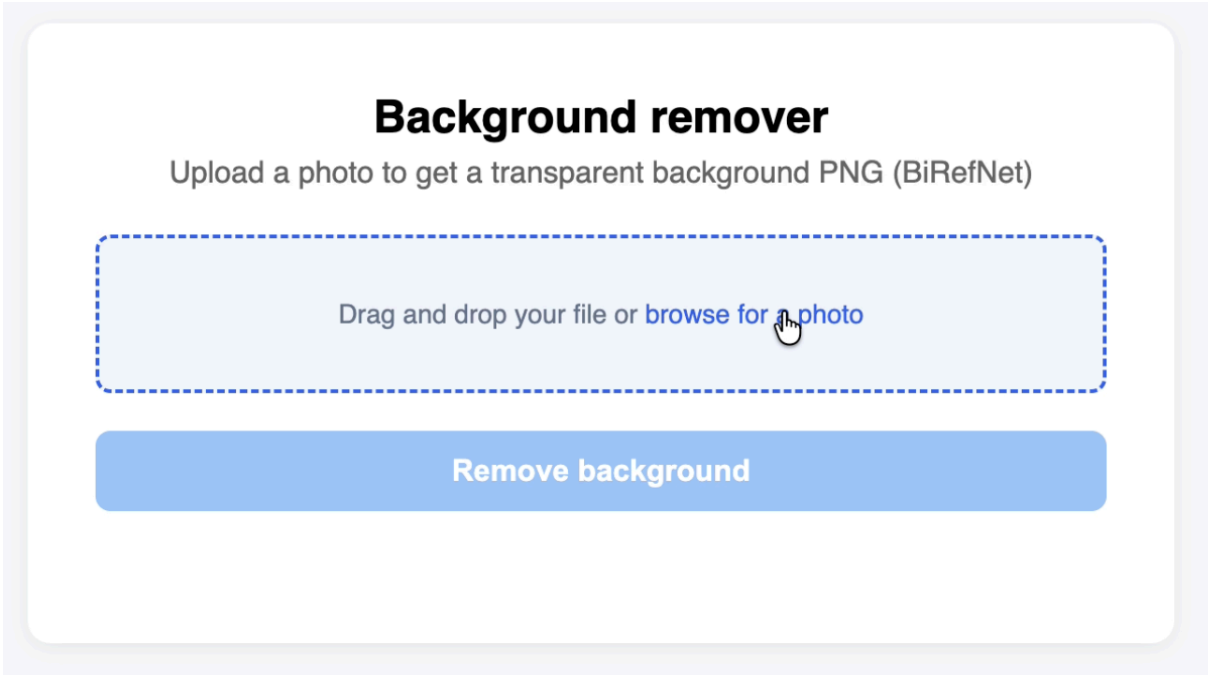
Eliminates ~3 sec delay on the first request; the model stays in memory between calls (~400 MB, loaded once at startup).

Request processing:

```
```python
img = Image.open(io.BytesIO(data)).convert("RGBA")
result = remove(img, session=session)
```
```

The service accepts a PIL image, performs inference using the initialized BiRefNet session, and returns an RGBA PNG image with an alpha channel.

Drag & drop support and checkerboard preview — visual transparency control without third-party dependencies (CSS repeating-conic-gradient).



4.3 API

| Method | Path | Description |
|--------|------------|--|
| GET | / | HTML-interface |
| POST | /remove-bg | Accepts multipart/form-data with file field, returns image/png |

5. Results

5.1 Segmentation Quality

Testing was conducted on portrait images with various background types:

| Background Type | MediaPipe Selfie | BiRefNet |
|------------------------------|----------------------|--------------|
| Uniform (studio) | Good | Excellent |
| Textured (wallpaper, nature) | Satisfactory | Excellent |
| Complex edges (hair) | Rough, strands merge | Detailed |
| Semi-transparent objects | Not supported | Satisfactory |



5.2 Performance on Apple M1 Pro

| Resolution | Processing Time | RAM Usage |
|-------------------|-----------------|-----------|
| 640×480 | ~0.45 s | ~1.2 GB |
| 1280×720 | ~0.65 s | ~1.4 GB |
| 1920×1080 | ~1.1 s | ~1.7 GB |
| 3840×2160
(4K) | ~3.2 s | ~2.8 GB |

Note: Performance values are approximate and are based on publicly available ONNX Runtime benchmarks together with practical testing on Apple Silicon devices. Actual performance may vary depending on image resolution and execution provider configuration. The model loads into memory at service startup (~3 s, one-time). Subsequent requests are processed without initialization delay.

6. Conclusion

0. A web service for background removal from portrait and product images was developed based on the BiRefNet model with a REST API and browser interface.

1. BiRefNet outperforms MediaPipe Selfie Segmentation in edge processing quality (hair, fine details) while maintaining acceptable speed for offline preprocessing tasks.

2. Implementation via ONNX Runtime + CoreML provides hardware acceleration on the Apple Silicon platform without CUDA dependency.

3. The developed solution demonstrates how modern open-source computer vision models can be rapidly integrated into production-ready web services for automated image preprocessing in e-commerce and related domains.